Zhanyuan Tian

🏠 Coniston Court, East Putney, London, UK

📫 zt1624@ic.ac.uk ☎ (+44) 07985066049 💻 GitHub: https://github.com/No-strum

📚 Education

Imperial College London **London, United Kingdom**
Doctor of Philosophy Sep 2024–Jun 2028
Skills: Fluid Mechanics, Numerical Simulation

Wuhan University **Hubei, China**
Bachelor of Engineering-Remote Sensing Sep 2020–Jun 2024
GPA: 3.92 92.01/100 (top 2%) **TOFEL**: 105 **GRE**: 330
Skills: Photogrammetry, Computer Vision, Machine Learning, Python, MATLAB, C++

🔧 Projects

Imperial College London – Hydrodynamics Laboratory London, UK
PhD in Fluid Mechanics 2024–Present
PhD Project: Automated Whitecap Detection, Tracking & Wave-Breaking Analysis

- Develop a computer-vision and remote-sensing framework to detect and track whitecaps in stereo observations of the sea surface, mapping image measurements into world coordinates to recover breaking-wave geometry and motion.
- Improve whitecap segmentation, background removal, and event tracking across varying sea states; estimate breaking-front velocity and propagation direction from image sequences for physically informed kinematic analysis.
- Fine-tune a BERT-based classifier using more than 10,000 annotated whitecaps to categorize breaking events and improve the reliability of downstream breaking-wave statistics.
- Connect image-derived whitecap statistics with wave kinematics, directional breaking distributions, and energy dissipation to advance the physical understanding of ocean-wave breaking.
- Tools: Python, PyTorch, OpenCV

CVRS Lab Hubei, China
Researcher May 2023–June 2024
Research Project:Depth-Prior MVSNet

- Optimized the network structure of Casmvsnet by employing a transformer for both intra- and inter-image feature fusion, leveraging the depth information from the previous stage to enhance the fused region in subsequent stages.
- Improved Cost Volume accuracy by implementing Bayesian estimation techniques: leveraged negative exponential powers of feature differences for likelihoods, utilized prior Cost Volume from previous stages, and employed 3D convolution for patch-based probability fusion after determining the depth posterior estimate for individual pixels.
- Utilized KL divergence to ensure a singular peak in the depth probability curve for each pixel, transforming the optimal depth estimation into a convex optimization problem. This enhancement increased the likelihood of network convergence.
- Outperformed other contemporary research studies and models during that period by gaining a result of **0.339** accuracy and **0.278** completeness metrics on DTU, indicating high levels of data accuracy and reliability.
- Tools: Python Pytorch Git

Research Project:SuperlightGlue Network

- Reconfigured the output layers of Superpoint network into a U-Net architecture, enabling simultaneous multi-scale descriptor extraction and matching ground truth calculations for each point.
- Augmented the SuperGlue network's efficacy with a pioneering coarse-to-fine aggregating technique that employed random sampling across three iterations. This method utilized transformers to merge multi-scale features of each point, identified high-confidence matches, and established them as prior information.
- Reduced runtime and GPU memory consumption on dense detection (**30.7%** and **12.0%** respectively) by limiting the attention aggregation range to the closest 10 well-matched point pairs while continuously iterating to obtain the final matching result.
- Tools: Python Pytorch Git

Assistant Researcher

- Partnered with a major global technology firm to develop a demonstration application, designed to fuse over 200 images from an iPad RGB-D camera into high-precision point clouds, utilizing open3D capabilities.
- Contributed to the enhancement of image fusion techniques by engineering the essential functionalities, including the implementation of homography transformation for a depth completion neural network.
- Evaluated the performance of established techniques, including Markov chain and bilateral filtering, for depth completion assignments, providing a comprehensive assessment of their strengths and limitations.
- Tools: Python(Open3d) Matlab

State Key Laboratory of Remote Sensing Sep 2022–Jun 2023

Project Leader

- Led a team of four in collecting 5.5km of road point cloud and image data, applying CSF for ground point cloud extraction, and constructing a road network model with the PTD (Progressive-densification-based filters) algorithm.
- Optimized remote sensing data in ground point clouds by augmenting information density using edge detection, elevating data quality through the removal of low-intensity points and attaining a 90% accuracy rate in road text recognition utilizing PaddleOCR and Densenet.
- Utilized MPR-GAN for image generation and SuperGlue for 2D image-level matching to optimize point rendering angles based on matches, completing multi-modal matching tasks and successfully calculating road camera position with **±10°** and **±15m** accuracy.
- Tools: C/C++ Python

Scholarships

National Scholarship of China Aug 2023
NITORI Scholarship Nov 2022
Haida Zhong Scholarship Sep 2021